

May-26-0482

CSPE-612(vi) (Soft Computing)

B.Tech, 6th (NEP)

Time : 3 Hours

Max. Marks : 60

Note : The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Section A, B, C, D. Section E is compulsory.

SECTION - A

1. (a) Define Soft Computing and list its features. **(4)**
(b) Explain the fundamentals of Artificial Neural Networks. **(4)**
(c) Explain the characteristics of neural networks. **(4)**

OR

2. (a) Define the neuron model. Explain the components of an artificial neuron. **(4)**
(b) Explain the Perceptron model. Discuss its features. **(4)**

SECTION - B

3. (a) Explain the Perceptron learning algorithm. **(4)**
(b) Write a short note on applications of ANN in the real world. **(4)**
(c) Explain different activation functions used in ANN. **(4)**

OR

4. (a) Explain the Multi-Layer Perceptron (MLP) with a neat diagram. **(4)**
(b) Explain the topology of a neural network architecture. **(4)**
(c) Explain the learning laws: Hebbian learning and Delta learning. **(4)**

SECTION - C

5. Explain the ART (Adaptive Resonance Theory) network model and its working. **(12)**

OR

6. Explain the Support Vector Machine (SVM) classifier. Describe ensemble methods in supervised learning. **(12)**

SECTION - D

7. Explain fuzzy models and the fuzzy rule base. Explain fuzzy inference methods (Mamdani and Sugeno). **(12)**

OR

8. Explain Neuro-Fuzzy systems and their advantages. Write applications of Neuro-Fuzzy systems. **(12)**

SECTION - E

(Compulsory)

9. (a) What is a perceptron?
(b) Define BAM.
(c) Define fuzzy relation.

- (d) What is a membership function?
 - (e) What is reinforcement learning?
 - (f) Define PCA. **(6×2=12)**
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Transcribed from a photograph of the original answer/question booklet. A few numeric values, table entries and one section could not be fully verified due to image quality (angle/blur/overlap with adjacent papers) — please cross-check against your original copy.